1 Project Management Basics class

Project Management Basics

Project Management Definition

Temporary endeavor undertaken to create a unique product, service or result. The temporary nature of projects indicates a definite beginning and end. The end is reached when the project h objectives have been achieved or when the project is terminated because its objectives will not or cannot be met, or when the need for the project no longer exists.
Source:
Project Management Body of Knowledge (PMBOK)

Project: "A temporary endeavor undertaken to create a unique product, service, or result."
Process: "A routine task repeated regularly."

Project concepts

Project Management: The application of knowledge, skills, tools, and techniques to project activities to meet project requirements
Project Manager: A person assigned by the performing organization to achieve the project objectives; accountable for the outcome of the project
Source:
Project Management Institute
PMBOK guide 6th and 7th edition

In practice:
Project is a special case of business operation.

The purpose of operations is to keep the organization functioning while the purpose of a project is to meet its goals and to conclude.

Therefore, operations are ongoing while projects are unique and temporary.

Project unique qualities vs process

- Temporary in nature (definite beginning and ending date)
- Parts of a project are scheduled.
- Project delivers something new or unique.

Goal of a project: On time and on budget while meeting objectives.

Value added by project management

- Reduce the time and money spent on a project.
- Anyone can complete a project if they have enough time and budget. Therefore, project management is about achieving the same results with the least amount of time and budget.
- Planning accounts for 80% of a project, while execution accounts for the remaining 20%.
- Stakeholder agreement is needed on the project's planned outcome as a list of deliverables with a fixed scope and quality.

Project Management concepts

- constraints
- key success factors
- project life cycle
- project process groups
- knowledge areas

Project Management planning pillars

- Skills needed (people)
- Knowhow needed (knowledge)
- Risks

- Enable constant communication

Product concepts

Product: A term used to describe all goods, services, and knowledge sold. Products are bundles of attributes (features, functions, benefits, and uses) and can be either tangible, as in the case of physical goods, or intangible, as in the case of those associated with service benefits, or a combination of the two.
Product Management: Ensuring over time that a product or service profitably meets the needs of customers by continually monitoring and modifying the elements of the marketing mix, including the product and its features, the communications strategy, distribution channels, and price.
Product Manager: The person assigned by the organization to have responsibility for overseeing all of the various activities concerning a particular product.
Source:
Product Development and Management Association

In practice:
A *product* is what you provide a group of users, whereas a *project* is a plan having a series of activities, a defined outcome, and fixed start and end dates.

Project managers own

- Scope
- Budget
- Delivery
- Resources
- Capacity
- Application of processes and knowledge
- Cross-team organization

- Problem resolution
- Risk management
- Customer engagement
- Status updates

Project manager vs Product manager

- Project manager focuses on "How" (work-oriented). Project managers guide key decisions to maximize quality and minimize risk.
- Product manager focuses on the "What" (functional requirements-oriented). Product managers guide key decisions to maximize value for customers and/or create new revenue streams.

Difference in Incentives

Project Manager - keep scope (no changes)
Product Manager - broaden scope (add new features)

Key deliverables of a Project Management

- Project Knowledge Base
- Project Portfolio Analysis
- Project Charter
- Project Management Plan
- Work Breakdown Structure
- Earned Value Analysis
- Feasibility Study
- Communication Plan
- Procurement Plan
- Quality Assurance Plan
- Risk Management Plan
- Status Reports
- Lesson's Learned

Project phases

1. **Define:** The project is discussed fully with all the stakeholders and the key objectives are identified. The costs and timescales are also established at this stage and there is often a feasibility study as well. This stage is complete when the project brief has been written and agreed.
2. **Plan:** An initial plan is developed. Planning is an ongoing activity because the plan is the basis for reviews and revision when necessary, depending on how the project progresses.
3. **Team formation:** Team assembly based on skill list of potential team members and their availability (time constraints).
4. **Communications:** should take place continuously, both within the project team and between the project team and stakeholders in the project, including anyone who contributes to achievement of the outcomes. Some communications will be through formal reporting procedures but many will be informal.
5. **Control:** Implementation takes place during the control stage. During this stage, the tasks and activities of the team will be monitored against the plan to assess the actual progress of the project against the planned progress. Control is essential to ensure that the objectives are met within the scheduled timescales, budgeted costs and quality. Regular reviews are usually held to enable the plan to be revised and for any difficulties that emerge to be resolved.
6. **Review and exit:** The review is held to evaluate whether all the intended outcomes of the project have been met. It is also important because it enables information to be gathered about the processes used in carrying out the project from which lessons can be learned for the future. The exit from the project has to be managed to ensure that:
    - any outstanding tasks are completed;

- all activities that were associated with the project are discontinued;
- all resources are accounted for, including any that remain at the end and have to be transferred or sold to someone else.

Project characteristics

- The purpose of a project is to meet its strategic goals and create a change in the organization to acquire new or additional tangible or intangible benefits that cannot be acquired by conducting daily operations.
- Projects produce unique products, services, or results.
- Project begins with a business case and project charter.

Project Constraints

- Enterprise Environmental Factors – EEFs: constraints, assumptions, and internal and external factors
- Iron triangle constrains: scope, schedule, cost
- Triple constraints: resources, quality, risks.
- Scope: the work performed to deliver a product, service, or result with the specified features and functions.

Project plan

Scope -> List of deliverables -> List of requirements -> List of activities -> List of activities in sequence -> People assignment to activities -> Net-gross time of activities from people responsible -> Project schedule (holidays included) -> Resource allocation -> Cost allocation -> Risks identification -> Quality Indicators -> Key Performance Indicators -> Stakeholder satisfaction feedback

Quality definition from ISO 9000:
"Quality is a delivered performance or result is the degree to which a set of inherent characteristics fulfill requirements."

Plan quality check:
Project plan from top -> down must converge from down -> top assessment.

Project risk assessment:
A risk is an uncertain event or condition that, if occurs, has a positive or negative effect on one or more project objectives. Negative risks are called threats, and positive risks are called opportunities. Two main factors determine the severity of a risk: (1) Probability of the occurrence, and (2) impact on the project. Risk responses are planned based on the severity level.

Main Project failure reasons:

- lack of clearly defined and/or achievable milestones and objectives to measure progress (37%),
- poor communication (19%),
- lack of communication by senior management (18%),
- employee resistance (14%),
- insufficient funding (9%).
  Source:
  PMI 2020 Pulse

Project life cycle

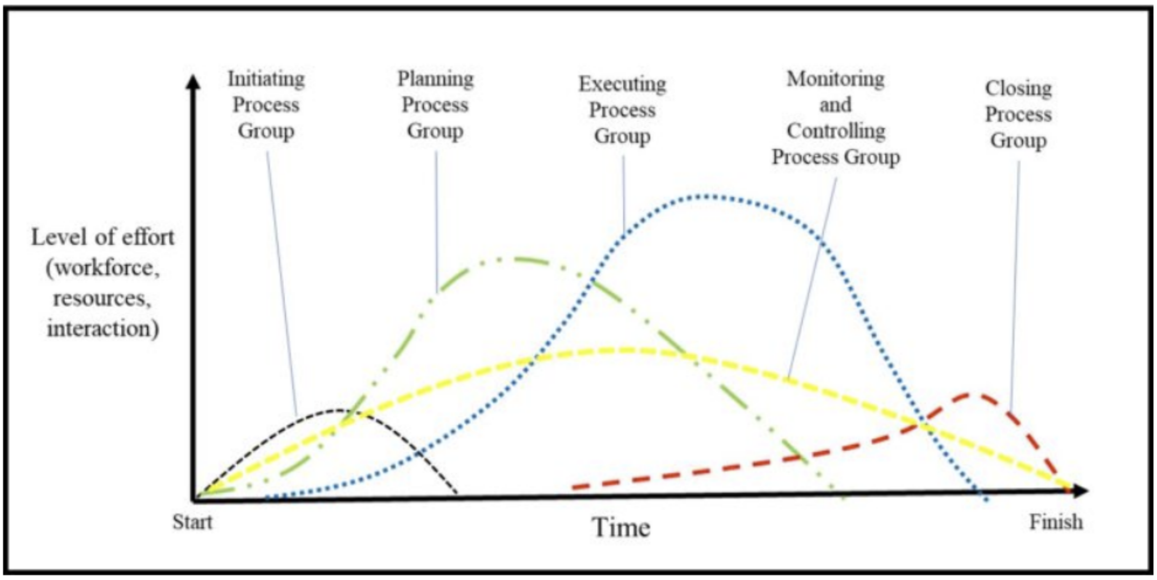
1. Initiating: to define a new project or a new phase of an existing project by obtaining authorization to start the project or phase.
   - Starting
   - Initiating
   - Conceptualizing
2. Executing: to complete the work defined in the project management plan to satisfy the project requirements.
   - Organizing and Preparing
   - Planning

- Analyzing and Designing

3. Monitoring and Controlling: to track, review, and regulate the progress and performance of the project, to identify any areas in which changes to the plan are required, and to initiate the corresponding changes.
    - Carrying Out the Work
    - Implementing
    - Executing
4. Closing: to formally complete or close the project, phase, or contract.
    - Ending the Project
    - Closing-out

    Source:

    PMBOK Guide 6th edition



Project Management areas

1. Project Integration Management
2. Project Scope Management
3. Project Schedule Management
4. Project Cost Management
5. Project Quality Management
6. Project Resource Management

7. Project Communications Management
8. Project Risk Management
9. Project Procurement Management
10. Project Stakeholder Management

Tools

Diagramming tools:

- Flowchart diagrams
- Sankey diagrams
- Version control (GitGraph) diagrams
- Gantt Charts (bar chart for project scheduling and resource assignment)
- PERT Charts (time dependencies between tasks and minimum time needed)

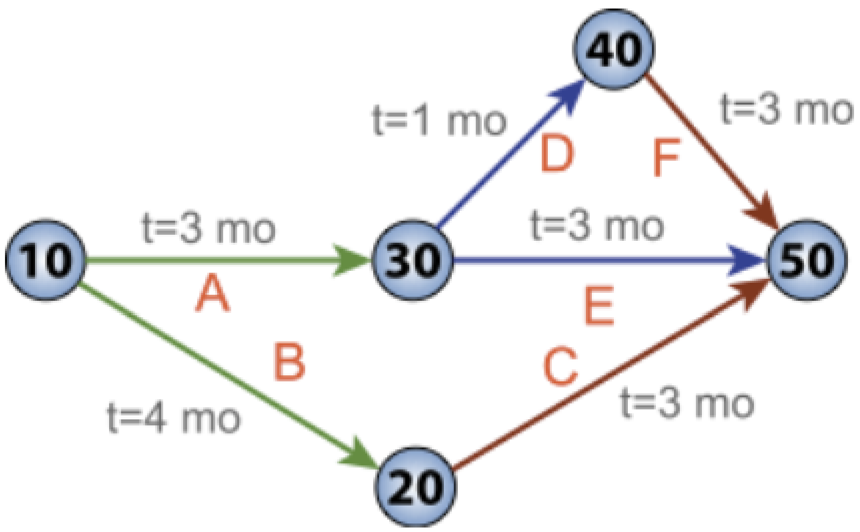


Figure 1.4: Pert Chart

- Critical Path Method (CPM) for determining schedule flexibility (each activity has optimistic and pessimistic start and end dates). Critical path is the longest full path of the project. Critical activities are the ones that must be finished first before the other can begin and if they delay the entire project will take longer.

- Business Process Model and Notation (BMPN) tools

Planning tools:

- Collaboration Technologies
- Critical Path Analysis
- PM software

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